

Stack Using Array

Aim:

To write a program to implement a stack using an array and perform basic operations.

Algorithm:

Step-1: Start Begin program execution.

Step-2: Initialize Stack – Define a stack of size MAX (5).

Step-3: Initialize top – Set top = -1 to indicate the stack is empty.

Step-4: Display Main Menu – Show options:

1. Push.
2. Pop.
3. Display.
4. Exit.

Step-5: Read User Choice – Read the choice from the user:

- If choice == 1, go to Push Operations Steps.
- If choice == 2, go to Pop Operations Steps.
- If choice == 3, go to Display Operations Steps.
- If choice == 4 , go to Exit Operations Steps.
- Otherwise, print “Invalid choice” and return to Step-4 (Display Main Menu).

Push() Operations Steps (Choice = 1):

Step-P1: Check Stack Overflow – If top == MAX-1, print “Stack Overflow” and return to Step-4 (Display Main Menu).

Step-P2: Read Item – Read the element to push from the user.

Step-P3: Increment top – $\text{top} = \text{top} + 1$.

Step-P4: Assign Stack Element – $\text{stack}[\text{top}] = \text{item}$.

Step-P5: Print Success Message – Print “Element pushed successfully” and return to Step-4 (Display Main Menu).

Pop() Operation Steps (Choice = 2):

Step-O1: Check Stack Underflow – If $\text{top} == -1$, print “Stack Underflow” and return to Step-4 (Display Main Menu).

Step-O2: Print Popped Element – Print $\text{stack}[\text{top}]$.

Step-O3: Decrement top – $\text{top} = \text{top} - 1$ and return to Step-4 (Display Main Menu).

Display() Operation Step (Choice = 3):

Step-D1: Check if Stack is Empty – If $\text{top} == -1$, print “Stack is empty” and return to Step-4 (Display Main Menu).

Step-D2: Loop through Stack – For $i = \text{top}$ down to 0, print $\text{stack}[i]$.

Step-D3: Return to Main Menu – Go back to Step-4 (Display Main Menu).

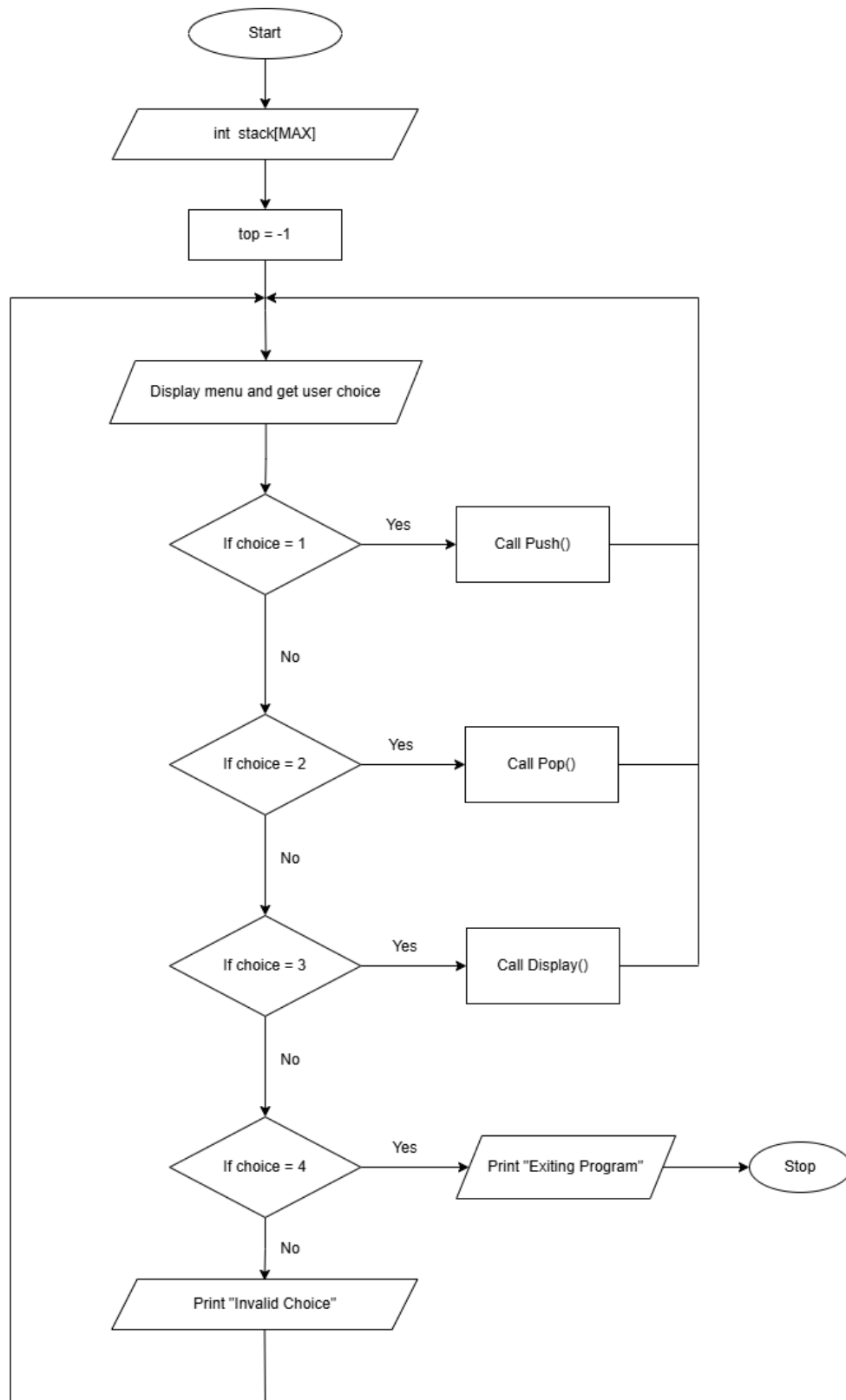
Exit() Operation Step (Choice = 4):

Step-E1: Print Exit Message – Print “Exiting program”.

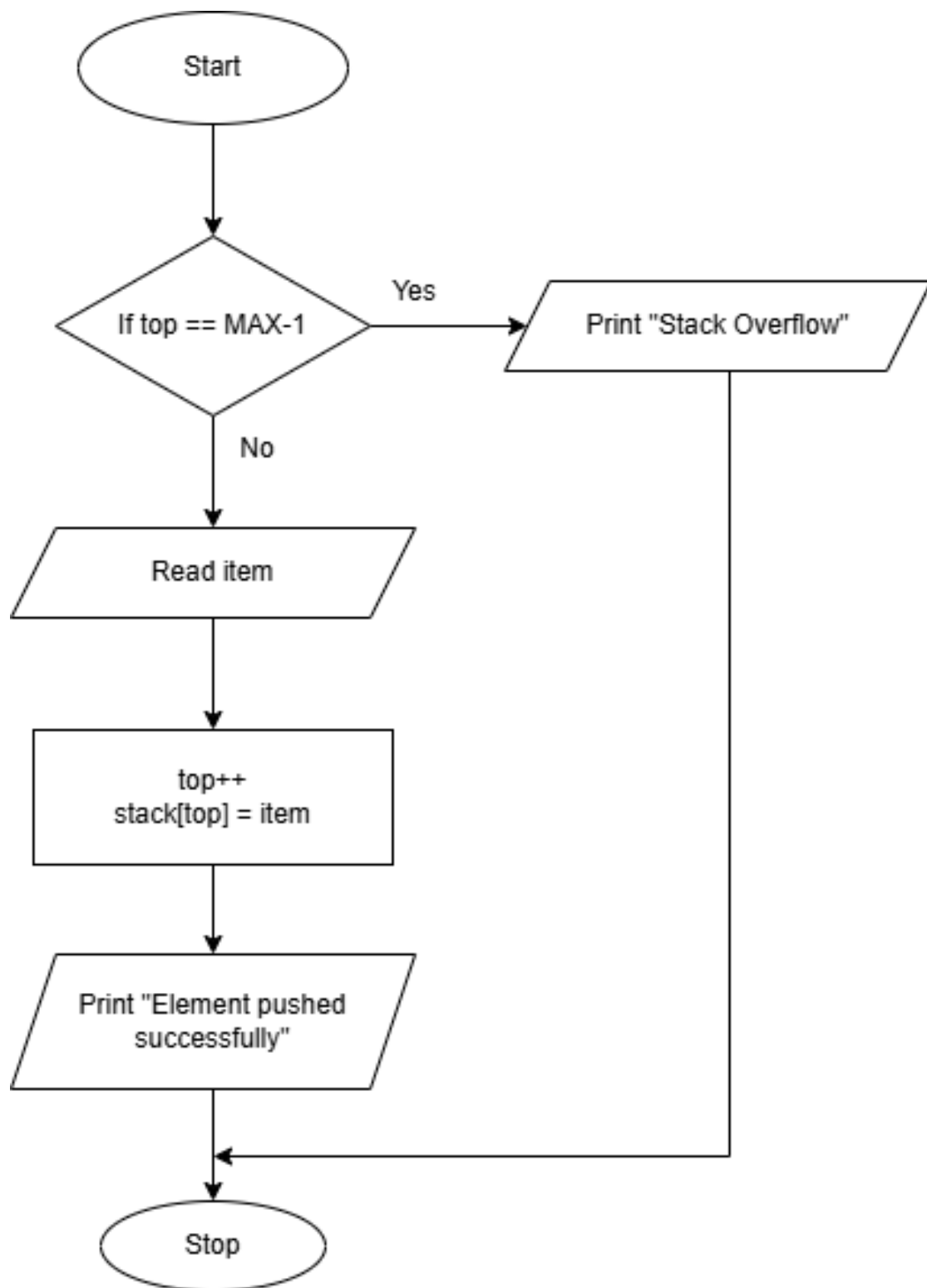
Step-E2: End – Program execution.

Flowchart:

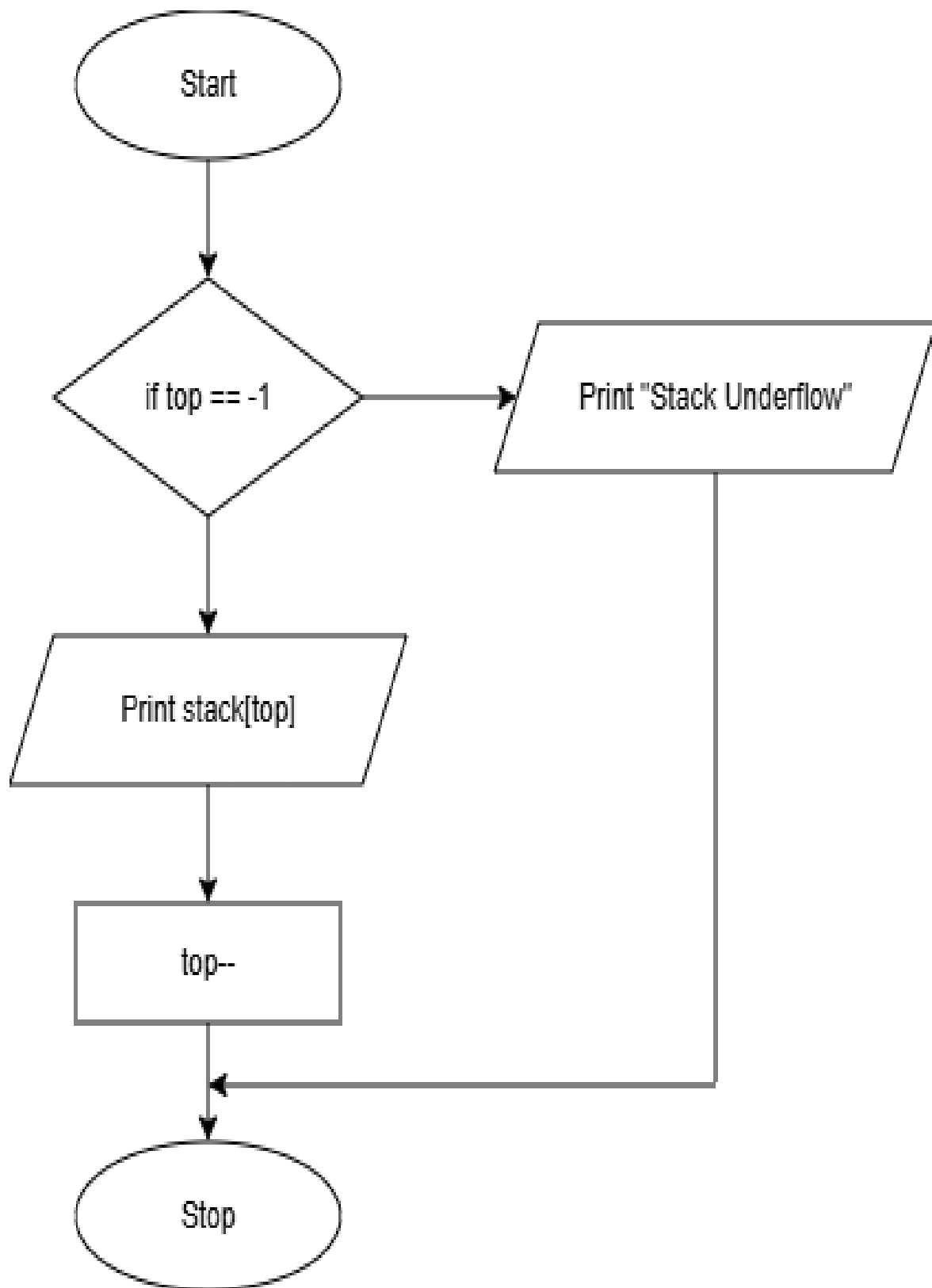
Main Menu:



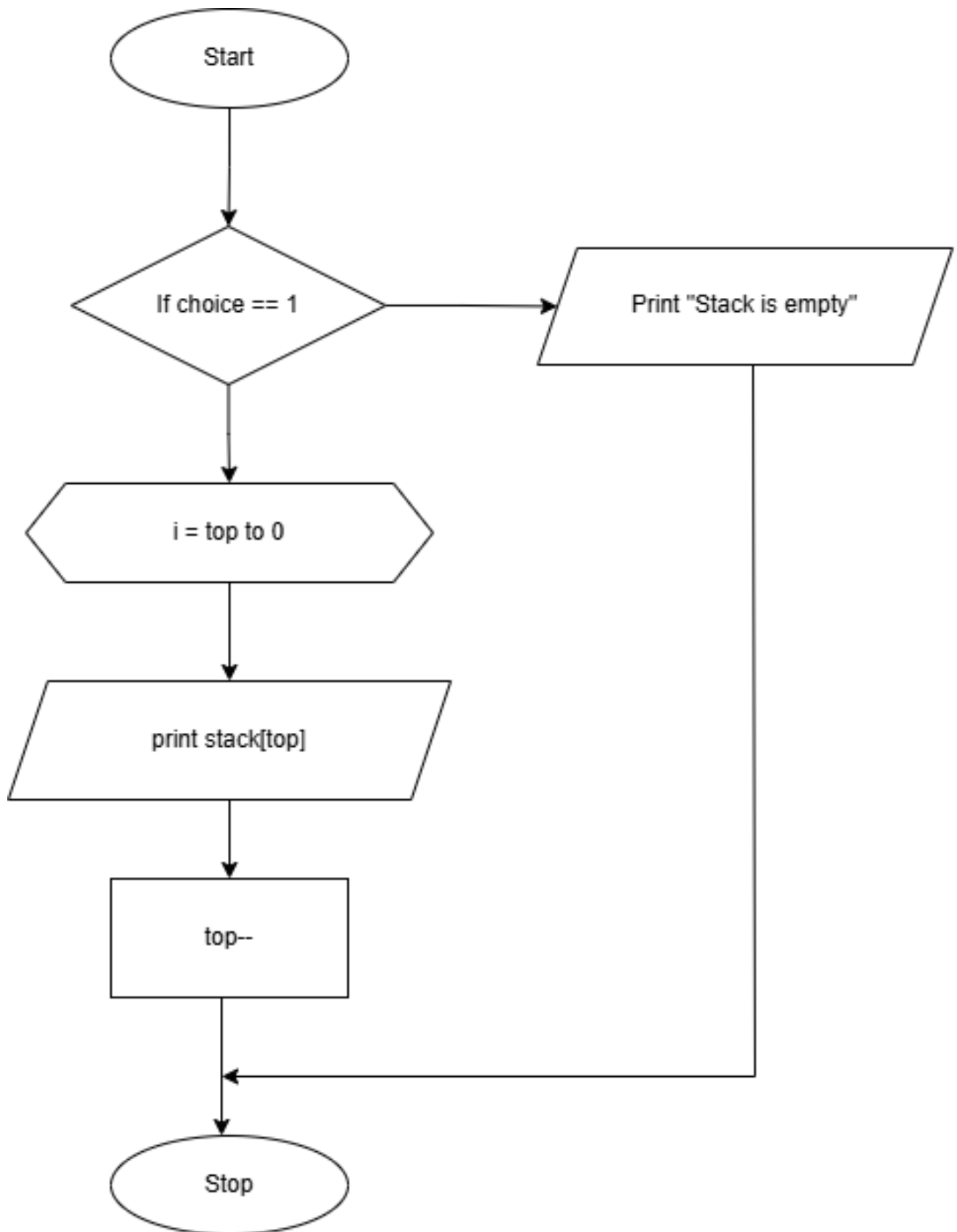
Push Operation:



Pop Operation:



Display Operation:



Source Code:

```
#include <stdio.h>

#define MAX 5

int stack[MAX], top = -1;

//Push operation
void push(){
    int item;
    if(top == MAX-1){
        printf("Stack Overflow!\n");
        return;
    }

    printf("Enter element to Push:");
    scanf("%d",&item);
    stack[++top] = item;
    printf("Element pushed successfully.\n");
}

//Pop operation
void pop(){
    if(top == -1){
        printf("Stack Underflow\n");
    }
}
```

```
        return;
    }
    printf("Popped element: %d\n", stack[top--]);
}
```

//Display Operation

```
void display(){
    if(top == -1){
        printf("Stack is empty.\n");
        return;
    }
    printf("Stack elements:\n");
    for(int i=top; i>=0; i--){
        printf("%d\n", stack[i]);
    }
}
```

```
int main(){
    int choice;

    do{
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n2. Pop\n3. Display\n4. Exit\n");
```



```
printf("Enter your choice: ");  
scanf("%d", &choice);  
  
switch(choice){  
    case 1: push(); break;  
    case 2: pop(); break;  
    case 3: display(); break;  
    case 4: printf("Exiting program.\n"); break;  
    default: printf("Invalid choice!\n");  
}  
}while(choice != 4);  
  
return 0;  
}
```

Output:

```
--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element to Push:10
Element pushed successfully.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element to Push:20
Element pushed successfully.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element to Push:30
Element pushed successfully.
```

```
--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack elements:
30
20
10

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2
Popped element: 30

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element to Push:40
Element pushed successfully.
```

```
--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack elements:
40
20
10

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 4
Exiting program.

...Program finished with exit code 0
Press ENTER to exit console.
```

Result:

Expected Output has been generated and have been verified.